Complete schematic - sheet 1

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

BT1 | 3S LiPo / connector

INVENTORY / VERIFY

11.1 V nominal / 3300 mAh

PACK+	PACK_RAW
PACK-	GND
BAL_CELL0	CELL0
BAL_CELL1	CELL1
BAL_CELL2	CELL2
BAL_CELL3	CELL3

TB_GND | Common ground distribution

ADDITIONAL / MISSING OR SELECT

Conductor / terminal rating TBD

BUS	GND

S2 | Latching E-stop / 3 isolated NC contacts

ADDITIONAL / MISSING OR SELECT

SELECT PART

COIL_COM	COIL_PG2
COIL_NC	COIL_STOP_CHAIN
SENSE_COM	V3V3
SENSE_NC	SENSE_STOP_OK
EN_COM	ENABLE_5V
EN_NC	EN_DRIVER_5V

Z_COIL | Coil suppression network

ADDITIONAL / MISSING OR SELECT

SELECT FOR K1

A1	COIL_HOLD
A2	GND

Z_L | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	LEFT_SUPPLY
-	GND

F0 | main pack fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_RAW
OUT	PACK_FUSED

P1 | 3S high-side protection assembly

ADDITIONAL / MISSING OR SELECT

SELECT PART

IN+	PACK_ISOLATED
OUT+	PACK_PROTECTED
GND	GND
CELL0	CELL0
CELL1	CELL1
CELL2	CELL2
CELL3	CELL3

S3 | Momentary motor-power ARM

ADDITIONAL / MISSING OR SELECT

NO; rating TBD

COM	COIL_STOP_CHAIN
NO	COIL_HOLD

OV1 | 5V rail 1 OVP/OCP and supervision assembly

ADDITIONAL / MISSING OR SELECT

SELECT / VERIFY

IN+	V5_MCU_RAW
OUT+	V5_MCU
GND	GND
PG_COM	COIL_FUSED
PG_NO	COIL_PG1

Z_R | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	RIGHT_SUPPLY
-	GND

S0 | Master load isolator

ADDITIONAL / MISSING OR SELECT

DC rating TBD

COM	PACK_FUSED
OUT	PACK_ISOLATED

TB_POS | Protected positive distribution

ADDITIONAL / MISSING OR SELECT

Conductor / terminal rating TBD

BUS	PACK_PROTECTED

K1 | DC motor contactor

ADDITIONAL / MISSING OR SELECT

CONTACT / COIL TBD

MAIN_COM	PACK_PROTECTED
MAIN_NO	MOTOR_SWITCHED
A1	COIL_HOLD
A2	GND
AUX_HOLD_COM	COIL_STOP_CHAIN
AUX_HOLD_NO	COIL_HOLD
AUX_SENSE_COM	SENSE_STOP_OK
AUX_SENSE_NO	ESTOP_OK

OV2 | 5V rail 2 OVP/OCP and supervision assembly

ADDITIONAL / MISSING OR SELECT

SELECT / VERIFY

IN+	V5_LOGIC_RAW
OUT+	V5_LOGIC
GND	GND
PG_COM	COIL_PG1
PG_NO	COIL_PG2

Z_A | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	AUX_SUPPLY
-	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 2

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

F1 | MCU buck input fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_PROTECTED
OUT	MCU_BUCK_IN

BUCK2 | LM2596 #2

INVENTORY / VERIFY

5.0 V target

IN+	LOGIC_BUCK_IN
IN-	GND
OUT+	V5_LOGIC_RAW
OUT-	GND

FA | auxiliary driver fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	MOTOR_SWITCHED
OUT	AUX_SUPPLY

BUCK1 | LM2596 #1

INVENTORY / VERIFY

5.0 V target

IN+	MCU_BUCK_IN
IN-	GND
OUT+	V5_MCU_RAW
OUT-	GND

FL | left driver fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	MOTOR_SWITCHED
OUT	LEFT_SUPPLY

F2 | logic buck input fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_PROTECTED
OUT	LOGIC_BUCK_IN

FK | contactor coil fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_PROTECTED
OUT	COIL_FUSED

FR | right driver fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	MOTOR_SWITCHED
OUT	RIGHT_SUPPLY

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 3

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

ESP1 | ESP32-WROOM-32 DevKit

INVENTORY / VERIFY

module terminal symbol

5V_VIN	V5_MCU
3V3	V3V3
GND	GND
GPIO16	L_FWD_3V3
GPIO17	L_REV_3V3
GPIO18	R_FWD_3V3
GPIO19	R_REV_3V3
GPIO14	ENABLE_3V3
GPIO21	BRUSH_IN1
GPIO22	BRUSH_IN2
GPIO23	CONVEYOR_IN3
GPIO25	CONVEYOR_IN4
GPIO32	TF_3V3
GPIO34	EF_SAFE
GPIO33	TL_3V3
GPIO35	EL_SAFE
GPIO13	TR_3V3
GPIO36	ER_SAFE
GPIO27	ESTOP_OK
GPIO0	NC - INSULATE
GPIO2	NC - INSULATE
GPIO4	NC - INSULATE
GPIO5	NC - INSULATE
GPIO12	NC - INSULATE
GPIO15	NC - INSULATE
GPIO26	NC - INSULATE
GPIO39	NC - INSULATE
UART0_TX_GPIO1	NC - INSULATE
UART0_RX_GPIO3	NC - INSULATE
EN_RESET	NC - INSULATE

US_L | HC-SR04 left

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TL_5V
ECHO	EL_RAW
GND	GND

U1 | SN74AHCT244N 3.3-to-5 V buffer

ADDITIONAL / MISSING OR SELECT

SN74AHCT244N

1 / 1OE_N	GND
2 / 1A1	L_FWD_3V3
3 / 2Y4	TR_5V
4 / 1A2	L_REV_3V3
5 / 2Y3	TL_5V
6 / 1A3	R_FWD_3V3
7 / 2Y2	TF_5V
8 / 1A4	R_REV_3V3
9 / 2Y1	ENABLE_5V
10 / GND	GND
11 / 2A1	ENABLE_3V3
12 / 1Y4	R_REV_5V
13 / 2A2	TF_3V3
14 / 1Y3	R_FWD_5V
15 / 2A3	TL_3V3
16 / 1Y2	L_REV_5V
17 / 2A4	TR_3V3
18 / 1Y1	L_FWD_5V
19 / 2OE_N	GND
20 / VCC	V5_LOGIC

DRV_A | ONE L298N: A brushes / B conveyor

INVENTORY / VERIFY

module terminal symbol

Vs_12V	AUX_SUPPLY
5V_LOGIC	V5_LOGIC
GND	GND
IN1	BRUSH_IN1
IN2	BRUSH_IN2
IN3	CONVEYOR_IN3
IN4	CONVEYOR_IN4
ENA	EN_DRIVER_5V
ENB	EN_DRIVER_5V
OUT1	DRV_A_OUT1
OUT2	DRV_A_OUT2
OUT3	DRV_A_OUT3
OUT4	DRV_A_OUT4

DRV_L | BTS7960 / IBT-2 left

INVENTORY / VERIFY

module terminal symbol

B+	LEFT_SUPPLY
B-	GND
M+	DRV_L_MP
M-	DRV_L_MN
VCC	V5_LOGIC
GND	GND
RPWM	L_FWD_5V
LPWM	L_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

DRV_R | BTS7960 / IBT-2 right

INVENTORY / VERIFY

module terminal symbol

B+	RIGHT_SUPPLY
B-	GND
M+	DRV_R_MP
M-	DRV_R_MN
VCC	V5_LOGIC
GND	GND
RPWM	R_FWD_5V
LPWM	R_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

US_F | HC-SR04 front

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TF_5V
ECHO	EF_RAW
GND	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 4

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

US_R | HC-SR04 right

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TR_5V
ECHO	ER_RAW
GND	GND

M5 | brush 1 motor

INVENTORY / VERIFY

12 V

A	M	M5_A
B		DRV_A_OUT2

CAM1 | ESP32-CAM - later phase

INVENTORY / VERIFY

reserved terminal symbol

5V	NC - INSULATE
GND	NC - INSULATE
U0R	NC - INSULATE
U0T	NC - INSULATE

FM3 | motor 3 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_R_MP
OUT	M3_A

FM6 | motor 6 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT1
OUT	M6_A

R24 | ECHO lower resistor

ADDITIONAL / MISSING OR SELECT

3.3k 1%, >=1/8 W

1	EF_DIV
2	GND

M1 | left front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M1_A
B		DRV_L_MN

M2 | left rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M2_A
B		DRV_L_MN

M3 | right front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M3_A
B		DRV_R_MN

M4 | right rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M4_A
B		DRV_R_MN

M6 | brush 2 motor

INVENTORY / VERIFY

12 V

A	M	M6_A
B		DRV_A_OUT2

M7 | conveyor motor

INVENTORY / VERIFY

Voltage TBD

A	M	M7_A
B		DRV_A_OUT4

FM1 | motor 1 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_L_MP
OUT	M1_A

FM2 | motor 2 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_L_MP
OUT	M2_A

FM4 | motor 4 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_R_MP
OUT	M4_A

FM5 | motor 5 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT1
OUT	M5_A

FM7 | motor 7 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT3
OUT	M7_A

R23 | ECHO upper resistor

ADDITIONAL / MISSING OR SELECT

2.2k 1%, >=1/8 W

1	EF_RAW
2	EF_DIV

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 5

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

R25 | ECHO upper resistor
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — EL_RAW
2 — EL_DIV

R26 | ECHO lower resistor
ADDITIONAL / MISSING OR SELECT
3.3k 1%, >=1/8 W
1 — EL_DIV
2 — GND

R27 | ECHO upper resistor
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — ER_RAW
2 — ER_DIV

R28 | ECHO lower resistor
ADDITIONAL / MISSING OR SELECT
3.3k 1%, >=1/8 W
1 — ER_DIV
2 — GND

R1 | L_FWD input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_FWD_3V3
2 — GND

R2 | L_REV input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_REV_3V3
2 — GND

R3 | R_FWD input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_FWD_3V3
2 — GND

R4 | R_REV input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_REV_3V3
2 — GND

R5 | ENABLE input pull-down
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — ENABLE_3V3
2 — GND

R6 | TF input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TF_3V3
2 — GND

R7 | TL input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TL_3V3
2 — GND

R8 | TR input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TR_3V3
2 — GND

R9 | BRUSH_IN1 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — BRUSH_IN1
2 — GND

R10 | BRUSH_IN2 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — BRUSH_IN2
2 — GND

R11 | CONVEYOR_IN3 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — CONVEYOR_IN3
2 — GND

R12 | CONVEYOR_IN4 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — CONVEYOR_IN4
2 — GND

R13 | DRV_L.R_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R14 | DRV_L.L_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 6

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

R15 | DRV_R.R_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R16 | DRV_R.L_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R17 | DRV_A.ENA local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R18 | DRV_A.ENB local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R19 | L_FWD receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_FWD_5V
2 — GND

R20 | L_REV receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_REV_5V
2 — GND

R21 | R_FWD receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_FWD_5V
2 — GND

R22 | R_REV receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_REV_5V
2 — GND

R29 | E-stop feedback pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — ESTOP_OK
2 — GND

C_U1 | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_F | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_L | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_R | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_BL | local motor-bus reservoir
ADDITIONAL / MISSING OR SELECT
C / ESR / voltage TBD
+ — LEFT_SUPPLY
- — GND

C_BR | local motor-bus reservoir
ADDITIONAL / MISSING OR SELECT
C / ESR / voltage TBD
+ — RIGHT_SUPPLY
- — GND

C_BA | local motor-bus reservoir
ADDITIONAL / MISSING OR SELECT
C / ESR / voltage TBD
+ — AUX_SUPPLY
- — GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 7

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

U2 | SN74LVC3G34DCU ECHO isolation

ADDITIONAL / MISSING OR SELECT

SN74LVC3G34DCU

1 / 1A	EF_DIV
2 / 3Y	ER_SAFE
3 / 2A	EL_DIV
4 / GND	GND
5 / 2Y	EL_SAFE
6 / 3A	ER_DIV
7 / 1Y	EF_SAFE
8 / VCC	V3V3

C_U2 | local ceramic bypass

ADDITIONAL / MISSING OR SELECT

100nF, >=10V

1	V3V3
2	GND

R30 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EF_SAFE
2	GND

R31 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EL_SAFE
2	GND

R32 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	ER_SAFE
2	GND

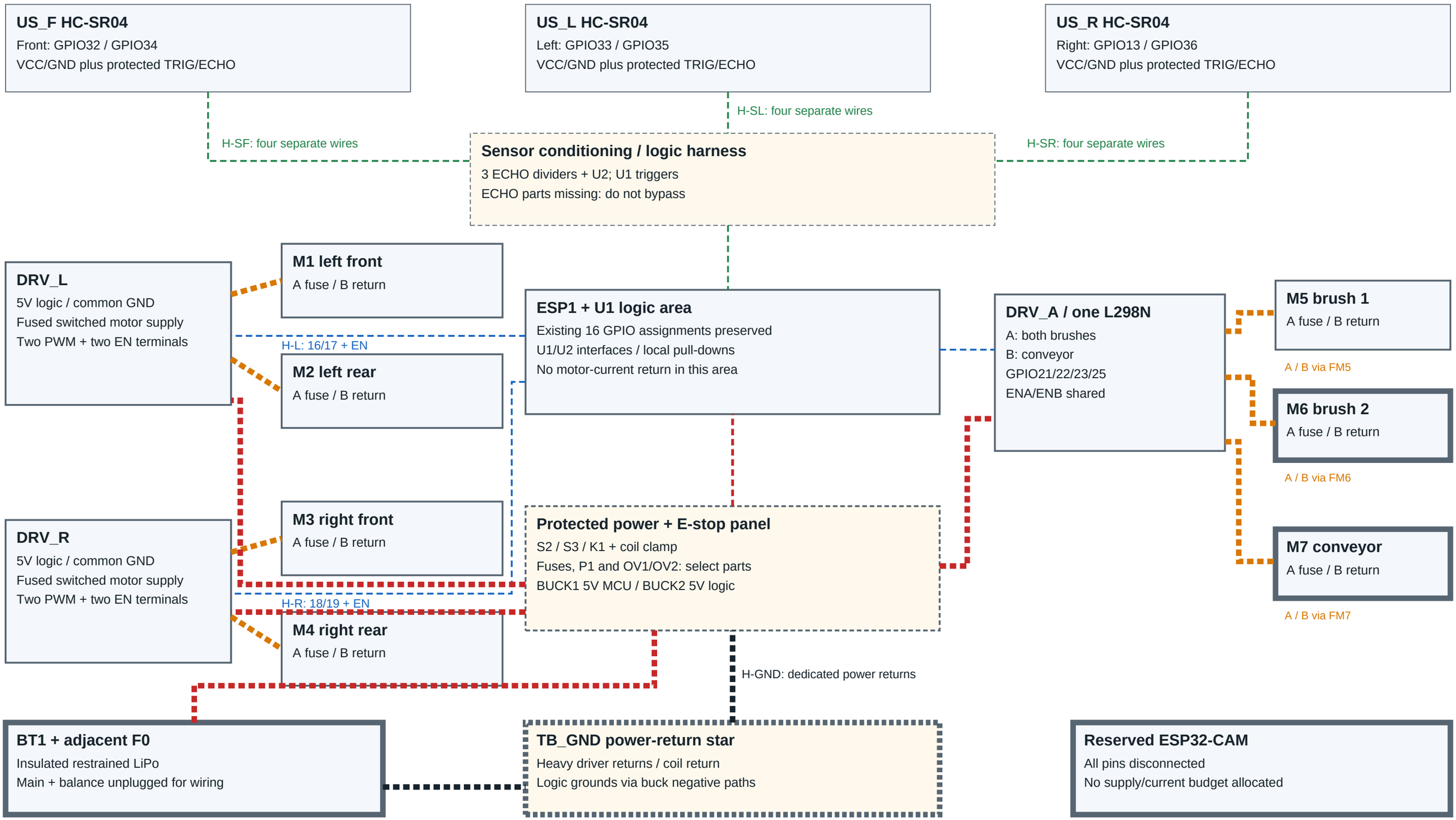
Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Physical harness layout - placement suggestion

Not to scale or connector orientation. Bundles are separate labeled conductors; use the full terminal/connection tables before fitting any wire.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



Keep high-current orange/red/black wiring separate from blue/green signal harnesses. Use insulated rated terminals; no breadboard motor paths. Exact wire gauge, length, connectors, routing and suppression remain release items.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

ESP32 pinout connection chart

Functional GPIO labels only. Exact DevKit header positions/USB power path remain unverified; no wiring orientation implied.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

GPIO / terminal	Firmware signal	Exact proposed endpoint chain	Electrical status / requirement
GPIO16	L_FWD	U1.2 -> U1.18 -> DRV_L.RPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO17	L_REV	U1.4 -> U1.16 -> DRV_L.LPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO18	R_FWD	U1.6 -> U1.14 -> DRV_R.RPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO19	R_REV	U1.8 -> U1.12 -> DRV_R.LPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO14	ENABLE	U1.11 -> U1.9 -> S2 EN NC -> six enables	**BLOCKED_MISSING** U1 + bias + cutoff missing
GPIO32	TF	U1.13 -> U1.7 -> US_F.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO33	TL	U1.15 -> U1.5 -> US_L.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO13	TR	U1.17 -> U1.3 -> US_R.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO21	BRUSH_IN1	DRV_A.IN1 / both brushes	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO22	BRUSH_IN2	DRV_A.IN2 / held LOW	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO23	CONVEYOR_IN3	DRV_A.IN3 / conveyor	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO25	CONVEYOR_IN4	DRV_A.IN4 / held LOW	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO34	EF	ECHO -> R23/R24 -> U2.1/7 -> GPIO34	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO35	EL	ECHO -> R25/R26 -> U2.3/5 -> GPIO35	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO36	ER	ECHO -> R27/R28 -> U2.6/2 -> GPIO36 / VP	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO27	ESTOP_OK	3V3 -> S2 NC -> K1 sense NO -> GPIO27	**BLOCKED_MISSING** Cutoff/feedback parts missing

5V/VIN: only V5_MCU after OV1 and actual-board qualification. 3V3: E-stop dry feedback + U2 only. GND: logic return to BUCK1 OUT-.
Never feed 5V into a GPIO or 3V3 pin.

Unused: GPIO26, GPIO39/VN, GPIO4. Reserve GPIO1/3 for serial, GPIO0/2/5/12/15 boot straps, GPIO6-11 flash. Keep EN/reset under onboard control; no JTAG on GPIO13/14. ESP32-CAM remains disconnected.

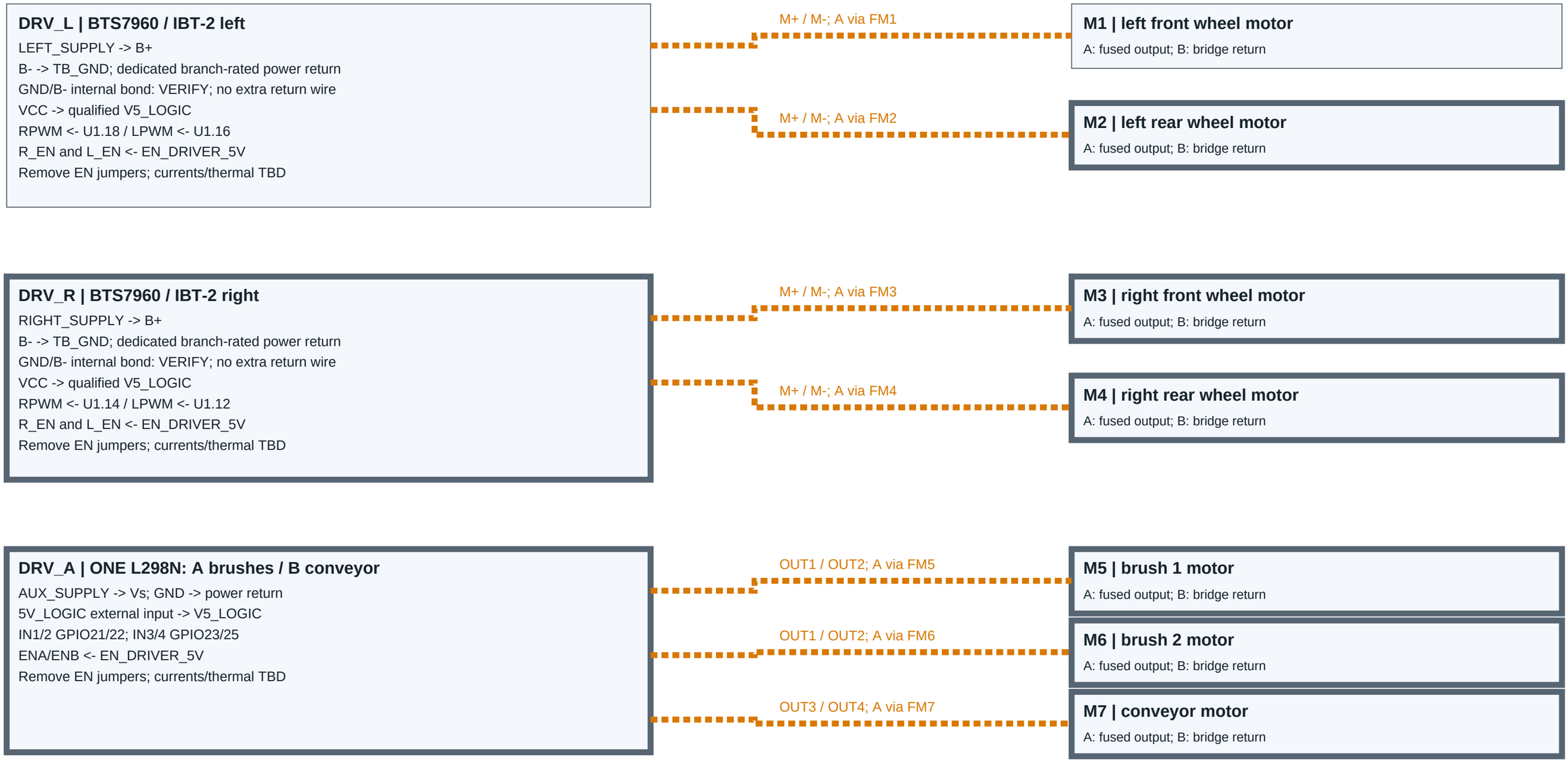
Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Motor-driver connections: seven motors

Each orange bundle represents two separate motor conductors. Exact per-terminal nets/fuses are shown in the complete schematic and CSV.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



M1/M2 and M3/M4 are parallel wheel pairs. M5/M6 are parallel on channel A; M7 is channel B. No motor lead to GND/chassis. Conveyor voltage and all motor running/start/stall currents remain unverified.

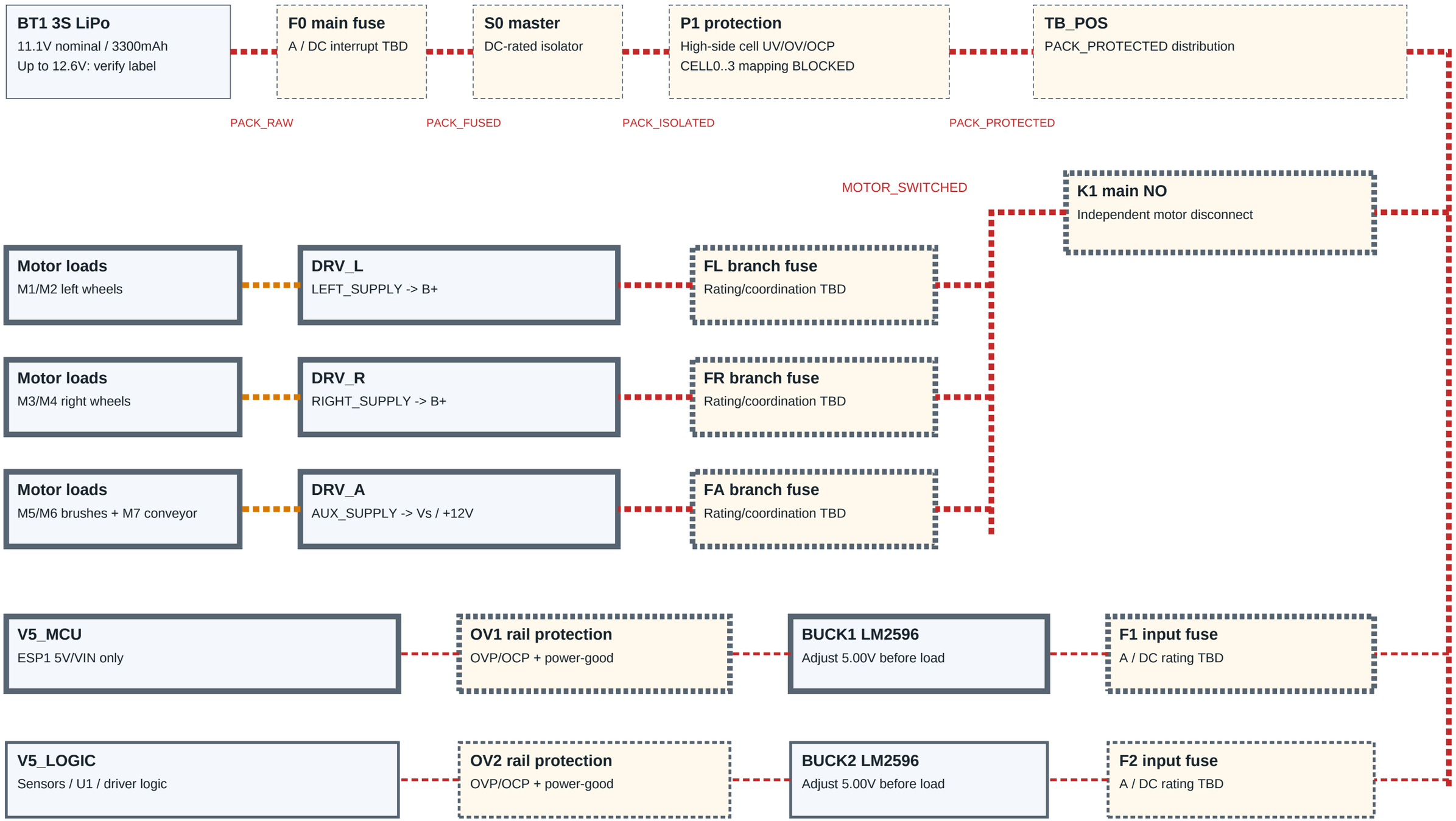
Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Battery and power distribution

High-current paths are thick red/black. All fuse, wire, coil, protection and motor-current ratings are TBD; no battery operation is approved.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



All returns -> TB_GND. Heavy driver returns; logic returns via buck OUT-. FK/coil chain: emergency_stop sheet. Do not parallel +5V outputs.

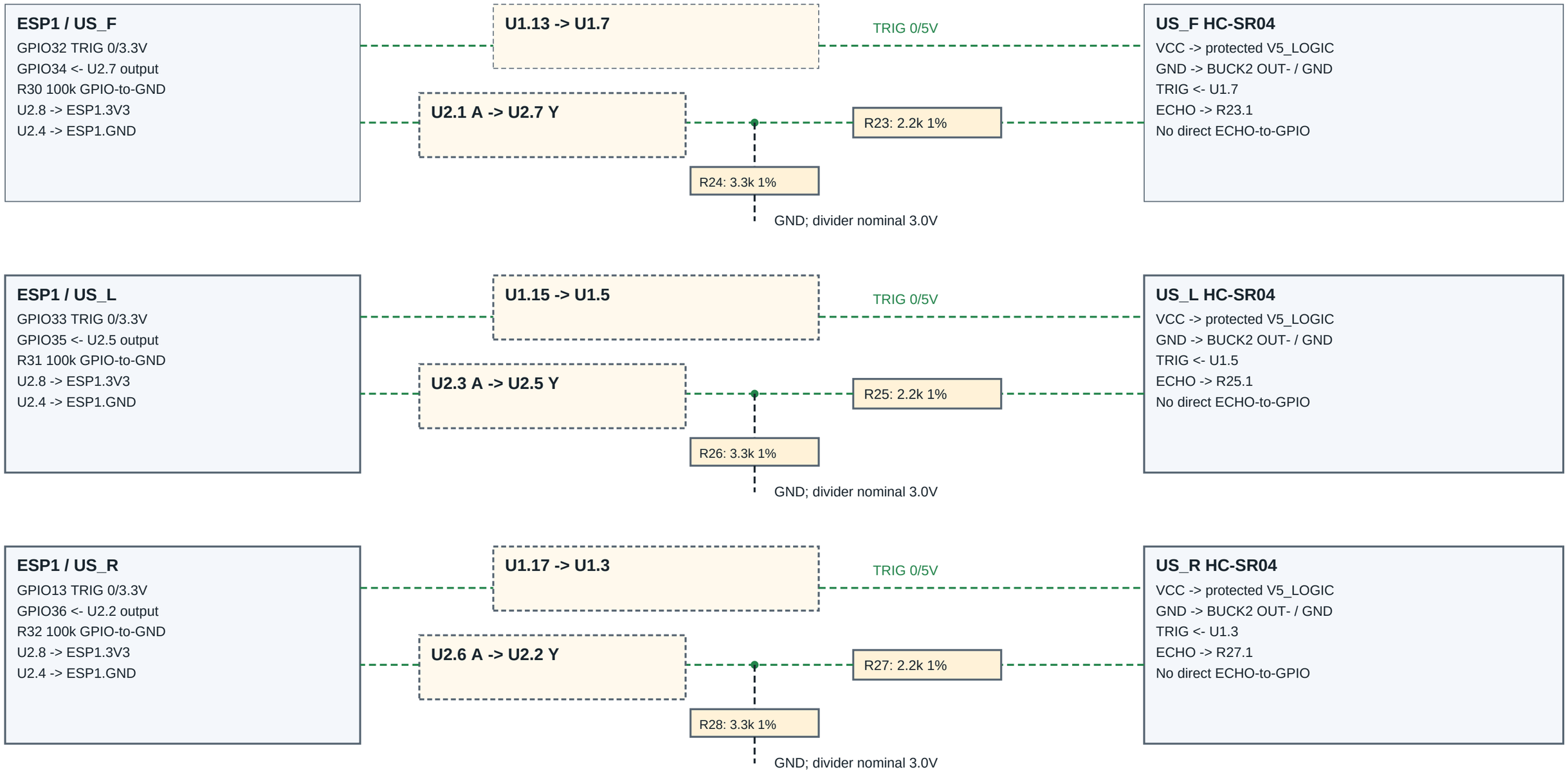
Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

HC-SR04: dividers plus powered-off ECHO isolation

U2 = SN74LVC3G34DCU; supply from MCU 3V3, not sensor 5V. Same GPIOs. Verify actual pulse levels/edges before connection.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



C_U2: 100nF U2.8-to-U2.4. U2 inputs: HIGH >=2.0V, LOW <=0.8V; edge rate <=10ns/V at 3.3V +/-0.3V. Keep divider-to-U2 traces short. Scope actual sensor cable and pulse shape. U2 outputs follow MCU supply; power-off Ioff limits backfeed.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

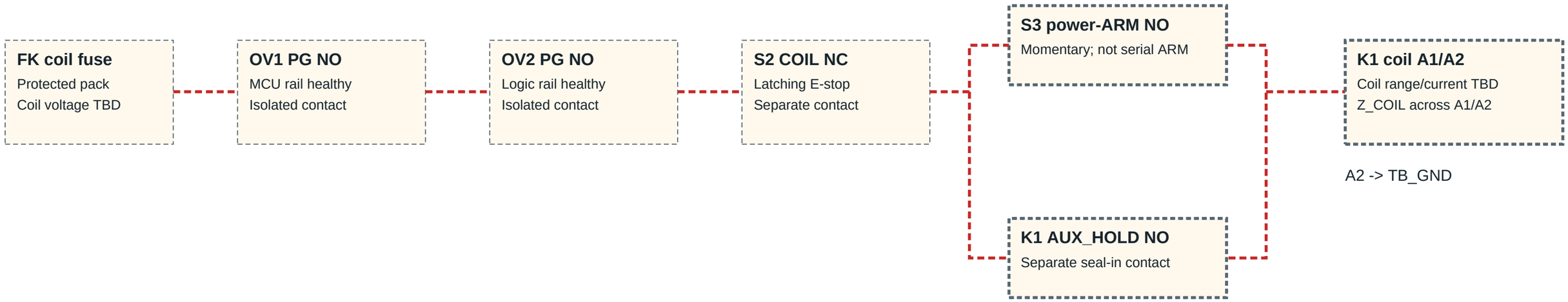
Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Independent emergency cutoff and manual power-arm

Functional contact roles only; actual contactor/supervisor/contact-block parts and terminal numbers must be selected and verified.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

1. Coil seal-in: independent of ESP32; S3 must be released for normal no-auto-restart behavior.



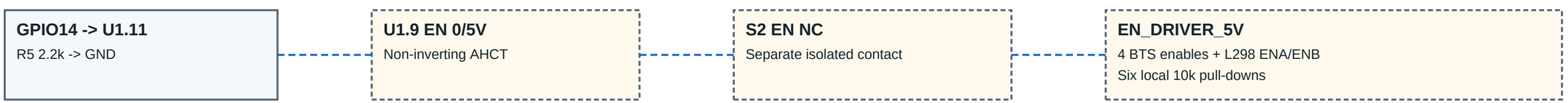
K1 MAIN_NO powers FL/FR/FA only. Stop opens coil + EN; K1 drops its seal-in and isolated feedback. Release alone does not restore power with S3 released. Held S3 / welded contacts are not covered by this functional single-channel circuit.

2. Firmware healthy input: HIGH only when E-stop released AND K1 closed.



K1-open boot intentionally latches ESTOP. Later: EN measured LOW -> deliberate power-ARM -> GPIO27 HIGH -> RESET -> ARM.

3. Shared enable: physical S2 contact opens every driver EN, independently of software.



Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Left BTS7960 and wheel pair

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

DRV_L | BTS7960 / IBT-2 left

INVENTORY / VERIFY

module terminal symbol

B+	LEFT_SUPPLY
B-	GND
M+	DRV_L_MP
M-	DRV_L_MN
VCC	V5_LOGIC
GND	GND
RPWM	L_FWD_5V
LPWM	L_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

M1 | left front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A		M1_A
B	M	DRV_L_MN

M2 | left rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A		M2_A
B	M	DRV_L_MN

FM1 | motor 1 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_L_MP
OUT		M1_A

FM2 | motor 2 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_L_MP
OUT		M2_A

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Right BTS7960 and wheel pair

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

DRV_R | BTS7960 / IBT-2 right

INVENTORY / VERIFY

module terminal symbol

B+	RIGHT_SUPPLY
B-	GND
M+	DRV_R_MP
M-	DRV_R_MN
VCC	V5_LOGIC
GND	GND
RPWM	R_FWD_5V
LPWM	R_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

M3 | right front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A — M3_A

B — DRV_R_MN

M4 | right rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A — M4_A

B — DRV_R_MN

FM3 | motor 3 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN — DRV_R_MP

OUT — M3_A

FM4 | motor 4 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN — DRV_R_MP

OUT — M4_A

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Conditional L298N and cleaning motors

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

DRV_A | ONE L298N: A brushes / B conveyor

INVENTORY / VERIFY

module terminal symbol

Vs_12V	AUX_SUPPLY
5V_LOGIC	V5_LOGIC
GND	GND
IN1	BRUSH_IN1
IN2	BRUSH_IN2
IN3	CONVEYOR_IN3
IN4	CONVEYOR_IN4
ENA	EN_DRIVER_5V
ENB	EN_DRIVER_5V
OUT1	DRV_A_OUT1
OUT2	DRV_A_OUT2
OUT3	DRV_A_OUT3
OUT4	DRV_A_OUT4

M5 | brush 1 motor

INVENTORY / VERIFY

12 V

A		M5_A
B	M	DRV_A_OUT2

M6 | brush 2 motor

INVENTORY / VERIFY

12 V

A		M6_A
B	M	DRV_A_OUT2

M7 | conveyor motor

INVENTORY / VERIFY

Voltage TBD

A		M7_A
B	M	DRV_A_OUT4

FM5 | motor 5 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_A_OUT1
OUT		M5_A

FM6 | motor 6 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_A_OUT1
OUT		M6_A

FM7 | motor 7 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_A_OUT3
OUT		M7_A

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

All seven motor terminals

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

M1 | left front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A ---- M1_A

M

B ---- DRV_L_MN

M2 | left rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A ---- M2_A

M

B ---- DRV_L_MN

M3 | right front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A ---- M3_A

M

B ---- DRV_R_MN

M4 | right rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A ---- M4_A

M

B ---- DRV_R_MN

M5 | brush 1 motor

INVENTORY / VERIFY

12 V

A ---- M5_A

M

B ---- DRV_A_OUT2

M6 | brush 2 motor

INVENTORY / VERIFY

12 V

A ---- M6_A

M

B ---- DRV_A_OUT2

M7 | conveyor motor

INVENTORY / VERIFY

Voltage TBD

A ---- M7_A

M

B ---- DRV_A_OUT4

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Buffers and default-low interface

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

U1 | SN74AHCT244N 3.3-to-5 V buffer

ADDITIONAL / MISSING OR SELECT

SN74AHCT244N

1 / 1OE_N	GND
2 / 1A1	L_FWD_3V3
3 / 2Y4	TR_5V
4 / 1A2	L_REV_3V3
5 / 2Y3	TL_5V
6 / 1A3	R_FWD_3V3
7 / 2Y2	TF_5V
8 / 1A4	R_REV_3V3
9 / 2Y1	ENABLE_5V
10 / GND	GND
11 / 2A1	ENABLE_3V3
12 / 1Y4	R_REV_5V
13 / 2A2	TF_3V3
14 / 1Y3	R_FWD_5V
15 / 2A3	TL_3V3
16 / 1Y2	L_REV_5V
17 / 2A4	TR_3V3
18 / 1Y1	L_FWD_5V
19 / 2OE_N	GND
20 / VCC	V5_LOGIC

R18 | DRV_A.ENB local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R32 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	ER_SAFE
2	GND

U2 | SN74LVC3G34DCU ECHO isolation

ADDITIONAL / MISSING OR SELECT

SN74LVC3G34DCU

1 / 1A	EF_DIV
2 / 3Y	ER_SAFE
3 / 2A	EL_DIV
4 / GND	GND
5 / 2Y	EL_SAFE
6 / 3A	ER_DIV
7 / 1Y	EF_SAFE
8 / VCC	V3V3

R14 | DRV_L.L_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R16 | DRV_R.L_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R30 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EF_SAFE
2	GND

R5 | ENABLE input pull-down

ADDITIONAL / MISSING OR SELECT

2.2k 1%, >=1/8 W

1	ENABLE_3V3
2	GND

R13 | DRV_L.R_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R15 | DRV_R.R_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R17 | DRV_A.ENA local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R31 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EL_SAFE
2	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Assembly: SOURCE PIN -> DESTINATION PIN / 1

Every canonical row is shown exactly once. INTERNAL_VERIFY is an existing module bond to inspect, not a wire to add.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W001	**BT1.PACK+ -> F0.IN** PACK_RAW | WIRE | Pack: up to 12.6 V	BLOCKED_RATING
W002	**F0.OUT -> S0.COM** PACK_FUSED | WIRE | Pack voltage	BLOCKED_RATING
W003	**S0.OUT -> P1.IN+** PACK_ISOLATED | WIRE | Pack voltage	BLOCKED_MISSING
W004	**P1.OUT+ -> TB_POS.BUS** PACK_PROTECTED | WIRE | Loaded pack voltage; up to 12.6 V	BLOCKED_MISSING
W005	**TB_POS.BUS -> K1.MAIN_COM** PACK_PROTECTED | WIRE | Loaded pack voltage; up to 12.6 V	BLOCKED_MISSING
W006	**TB_POS.BUS -> F1.IN** PACK_PROTECTED | WIRE | Loaded pack voltage; up to 12.6 V	BLOCKED_MISSING
W007	**TB_POS.BUS -> F2.IN** PACK_PROTECTED | WIRE | Loaded pack voltage; up to 12.6 V	BLOCKED_MISSING
W008	**TB_POS.BUS -> FK.IN** PACK_PROTECTED | WIRE | Loaded pack voltage; up to 12.6 V	BLOCKED_MISSING
W009	**K1.MAIN_NO -> FL.IN** MOTOR_SWITCHED | WIRE | Pack voltage when K1 closed	BLOCKED_RATING
W010	**K1.MAIN_NO -> FR.IN** MOTOR_SWITCHED | WIRE | Pack voltage when K1 closed	BLOCKED_RATING
W011	**K1.MAIN_NO -> FA.IN** MOTOR_SWITCHED | WIRE | Pack voltage when K1 closed	BLOCKED_RATING
W012	**FL.OUT -> DRV_L.B+** LEFT_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W013	**FL.OUT -> C_BL.+** LEFT_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W014	**FL.OUT -> Z_L.+** LEFT_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W015	**FR.OUT -> DRV_R.B+** RIGHT_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W016	**FR.OUT -> C_BR.+** RIGHT_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W017	**FR.OUT -> Z_R.+** RIGHT_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W018	**FA.OUT -> DRV_A.Vs_12V** AUX_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W019	**FA.OUT -> C_BA.+** AUX_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W020	**FA.OUT -> Z_A.+** AUX_SUPPLY | WIRE | Pack voltage	BLOCKED_RATING
W021	**F1.OUT -> BUCK1.IN+** MCU_BUCK_IN | WIRE | Pack voltage	BLOCKED_RATING
W022	**F2.OUT -> BUCK2.IN+** LOGIC_BUCK_IN | WIRE | Pack voltage	BLOCKED_RATING

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Assembly: SOURCE PIN -> DESTINATION PIN / 2

Every canonical row is shown exactly once. INTERNAL_VERIFY is an existing module bond to inspect, not a wire to add.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W023	**OV1.OUT+ -> ESP1.5V_VIN** V5_MCU | WIRE | Regulated 5.0 V target	BLOCKED_MISSING
W024	**OV2.OUT+ -> U1.20 (VCC)** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W025	**OV2.OUT+ -> DRV_L.VCC** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W026	**OV2.OUT+ -> DRV_R.VCC** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W027	**OV2.OUT+ -> DRV_A.5V_LOGIC** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W028	**OV2.OUT+ -> US_F.VCC** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W029	**OV2.OUT+ -> US_L.VCC** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W030	**OV2.OUT+ -> US_R.VCC** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W031	**OV2.OUT+ -> C_U1.1** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W032	**OV2.OUT+ -> C_F.1** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W033	**OV2.OUT+ -> C_L.1** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W034	**OV2.OUT+ -> C_R.1** V5_LOGIC | WIRE | Regulated 5.0 V target; qualify 4.75-5.25 V	BLOCKED_MISSING
W035	**ESP1.3V3 -> S2.SENSE_COM** V3V3 | WIRE | ESP32 3.3 V	PROVISIONAL
W036	**ESP1.3V3 -> U2.8 (VCC)** V3V3 | WIRE | ESP32 3.3 V	PROVISIONAL
W037	**ESP1.3V3 -> C_U2.1** V3V3 | WIRE | ESP32 3.3 V	PROVISIONAL
W038	**TB_GND.BUS -> BT1.PACK-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W039	**TB_GND.BUS -> P1.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W040	**BUCK1.OUT- -> ESP1.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W041	**TB_GND.BUS -> BUCK1.IN-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W042	**BUCK1.IN- -> BUCK1.OUT-** GND | INTERNAL_VERIFY | 0 V reference; returns sized by branch	PROVISIONAL
W043	**TB_GND.BUS -> BUCK2.IN-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W044	**BUCK2.IN- -> BUCK2.OUT-** GND | INTERNAL_VERIFY | 0 V reference; returns sized by branch	PROVISIONAL

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Assembly: SOURCE PIN -> DESTINATION PIN / 3

Every canonical row is shown exactly once. INTERNAL_VERIFY is an existing module bond to inspect, not a wire to add.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W045	**TB_GND.BUS -> DRV_L.B-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W046	**DRV_L.B- -> DRV_L.GND** GND | INTERNAL_VERIFY | 0 V reference; returns sized by branch	PROVISIONAL
W047	**TB_GND.BUS -> DRV_R.B-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W048	**DRV_R.B- -> DRV_R.GND** GND | INTERNAL_VERIFY | 0 V reference; returns sized by branch	PROVISIONAL
W049	**TB_GND.BUS -> DRV_A.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W050	**BUCK2.OUT- -> US_F.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W051	**BUCK2.OUT- -> US_L.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W052	**BUCK2.OUT- -> US_R.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W053	**BUCK2.OUT- -> U1.10 (GND)** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W054	**U1.10 (GND) -> U1.1 (1OE_N)** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W055	**U1.10 (GND) -> U1.19 (2OE_N)** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W056	**TB_GND.BUS -> K1.A2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W057	**K1.A2 -> Z_COIL.A2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W058	**ESP1.GND -> R24.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W059	**ESP1.GND -> R26.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W060	**ESP1.GND -> R28.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W061	**U1.10 (GND) -> R1.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W062	**U1.10 (GND) -> R2.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W063	**U1.10 (GND) -> R3.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W064	**U1.10 (GND) -> R4.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W065	**U1.10 (GND) -> R5.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W066	**U1.10 (GND) -> R6.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Assembly: SOURCE PIN -> DESTINATION PIN / 4

Every canonical row is shown exactly once. INTERNAL_VERIFY is an existing module bond to inspect, not a wire to add.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W067	**U1.10 (GND) -> R7.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W068	**U1.10 (GND) -> R8.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W069	**DRV_A.GND -> R9.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W070	**DRV_A.GND -> R10.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W071	**DRV_A.GND -> R11.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W072	**DRV_A.GND -> R12.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W073	**DRV_L.GND -> R13.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W074	**DRV_L.GND -> R14.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W075	**DRV_R.GND -> R15.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W076	**DRV_R.GND -> R16.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W077	**DRV_A.GND -> R17.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W078	**DRV_A.GND -> R18.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W079	**DRV_L.GND -> R19.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W080	**DRV_L.GND -> R20.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W081	**DRV_R.GND -> R21.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W082	**DRV_R.GND -> R22.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W083	**ESP1.GND -> R29.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W084	**U1.10 (GND) -> C_U1.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W085	**US_F.GND -> C_F.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W086	**US_L.GND -> C_L.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W087	**US_R.GND -> C_R.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W088	**DRV_L.B- -> C_BL.-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Assembly: SOURCE PIN -> DESTINATION PIN / 5

Every canonical row is shown exactly once. INTERNAL_VERIFY is an existing module bond to inspect, not a wire to add.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W089	**DRV_R.B- -> C_BR.-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W090	**DRV_A.GND -> C_BA.-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W091	**BUCK1.OUT- -> OV1.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W092	**BUCK2.OUT- -> OV2.GND** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W093	**DRV_L.B- -> Z_L.-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W094	**DRV_R.B- -> Z_R.-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W095	**DRV_A.GND -> Z_A.-** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W096	**ESP1.GND -> U2.4 (GND)** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W097	**U2.4 (GND) -> C_U2.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W098	**ESP1.GND -> R30.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W099	**ESP1.GND -> R31.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W100	**ESP1.GND -> R32.2** GND | WIRE | 0 V reference; returns sized by branch	PROVISIONAL
W101	**BT1.BAL_CELL0 -> P1.CELL0** CELL0 | WIRE | Cell-stack potential; order TBD	BLOCKED_RATING
W102	**BT1.BAL_CELL1 -> P1.CELL1** CELL1 | WIRE | Cell-stack potential; order TBD	BLOCKED_RATING
W103	**BT1.BAL_CELL2 -> P1.CELL2** CELL2 | WIRE | Cell-stack potential; order TBD	BLOCKED_RATING
W104	**BT1.BAL_CELL3 -> P1.CELL3** CELL3 | WIRE | Cell-stack potential; order TBD	BLOCKED_RATING
W105	**FK.OUT -> OV1.PG_COM** COIL_FUSED | WIRE | Pack voltage; K1 coil compatibility TBD	BLOCKED_RATING
W106	**S2.COIL_NC -> S3.COM** COIL_STOP_CHAIN | WIRE | Coil supply when stop released	BLOCKED_MISSING
W107	**S2.COIL_NC -> K1.AUX_HOLD_COM** COIL_STOP_CHAIN | WIRE | Coil supply when stop released	BLOCKED_MISSING
W108	**S3.NO -> K1.AUX_HOLD_NO** COIL_HOLD | WIRE | Coil supply when manually latched	BLOCKED_MISSING
W109	**S3.NO -> K1.A1** COIL_HOLD | WIRE | Coil supply when manually latched	BLOCKED_MISSING
W110	**S3.NO -> Z_COIL.A1** COIL_HOLD | WIRE | Coil supply when manually latched	BLOCKED_MISSING

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Assembly: SOURCE PIN -> DESTINATION PIN / 6

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E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W111	**K1.AUX_SENSE_NO -> ESP1.GPIO27** ESTOP_OK | WIRE | Healthy 3.3 V; open/pressed 0 V	BLOCKED_MISSING
W112	**K1.AUX_SENSE_NO -> R29.1** ESTOP_OK | WIRE | Healthy 3.3 V; open/pressed 0 V	BLOCKED_MISSING
W113	**ESP1.GPIO16 -> U1.2 (1A1)** L_FWD_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W114	**ESP1.GPIO16 -> R1.1** L_FWD_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W115	**U1.18 (1Y1) -> DRV_L.RPWM** L_FWD_5V | WIRE | 0/5 V	BLOCKED_MISSING
W116	**U1.18 (1Y1) -> R19.1** L_FWD_5V | WIRE | 0/5 V	BLOCKED_MISSING
W117	**ESP1.GPIO17 -> U1.4 (1A2)** L_REV_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W118	**ESP1.GPIO17 -> R2.1** L_REV_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W119	**U1.16 (1Y2) -> DRV_L.LPWM** L_REV_5V | WIRE | 0/5 V	BLOCKED_MISSING
W120	**U1.16 (1Y2) -> R20.1** L_REV_5V | WIRE | 0/5 V	BLOCKED_MISSING
W121	**ESP1.GPIO18 -> U1.6 (1A3)** R_FWD_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W122	**ESP1.GPIO18 -> R3.1** R_FWD_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W123	**U1.14 (1Y3) -> DRV_R.RPWM** R_FWD_5V | WIRE | 0/5 V	BLOCKED_MISSING
W124	**U1.14 (1Y3) -> R21.1** R_FWD_5V | WIRE | 0/5 V	BLOCKED_MISSING
W125	**ESP1.GPIO19 -> U1.8 (1A4)** R_REV_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W126	**ESP1.GPIO19 -> R4.1** R_REV_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W127	**U1.12 (1Y4) -> DRV_R.LPWM** R_REV_5V | WIRE | 0/5 V	BLOCKED_MISSING
W128	**U1.12 (1Y4) -> R22.1** R_REV_5V | WIRE | 0/5 V	BLOCKED_MISSING
W129	**ESP1.GPIO14 -> U1.11 (2A1)** ENABLE_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W130	**ESP1.GPIO14 -> R5.1** ENABLE_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W131	**U1.9 (2Y1) -> S2.EN_COM** ENABLE_5V | WIRE | 0/5 V	BLOCKED_MISSING
W132	**ESP1.GPIO32 -> U1.13 (2A2)** TF_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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Assembly: SOURCE PIN -> DESTINATION PIN / 7

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E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W133	**ESP1.GPIO32 -> R6.1** TF_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W134	**U1.7 (2Y2) -> US_F.TRIG** TF_5V | WIRE | 0/5 V	BLOCKED_MISSING
W135	**ESP1.GPIO33 -> U1.15 (2A3)** TL_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W136	**ESP1.GPIO33 -> R7.1** TL_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W137	**U1.5 (2Y3) -> US_L.TRIG** TL_5V | WIRE | 0/5 V	BLOCKED_MISSING
W138	**ESP1.GPIO13 -> U1.17 (2A4)** TR_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W139	**ESP1.GPIO13 -> R8.1** TR_3V3 | WIRE | 0/3.3 V	BLOCKED_MISSING
W140	**U1.3 (2Y4) -> US_R.TRIG** TR_5V | WIRE | 0/5 V	BLOCKED_MISSING
W141	**S2.EN_NC -> DRV_L.R_EN** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W142	**S2.EN_NC -> DRV_L.L_EN** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W143	**S2.EN_NC -> DRV_R.R_EN** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W144	**S2.EN_NC -> DRV_R.L_EN** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W145	**S2.EN_NC -> DRV_A.ENA** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W146	**S2.EN_NC -> DRV_A.ENB** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W147	**S2.EN_NC -> R13.1** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W148	**S2.EN_NC -> R14.1** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W149	**S2.EN_NC -> R15.1** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W150	**S2.EN_NC -> R16.1** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W151	**S2.EN_NC -> R17.1** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W152	**S2.EN_NC -> R18.1** EN_DRIVER_5V | WIRE | 0/5 V; LOW disabled	BLOCKED_MISSING
W153	**ESP1.GPIO21 -> DRV_A.IN1** BRUSH_IN1 | WIRE | 0/3.3 V	PROVISIONAL
W154	**ESP1.GPIO21 -> R9.1** BRUSH_IN1 | WIRE | 0/3.3 V	PROVISIONAL

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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Assembly: SOURCE PIN -> DESTINATION PIN / 8

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E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W155	**ESP1.GPIO22 -> DRV_A.IN2** BRUSH_IN2 | WIRE | 0/3.3 V	PROVISIONAL
W156	**ESP1.GPIO22 -> R10.1** BRUSH_IN2 | WIRE | 0/3.3 V	PROVISIONAL
W157	**ESP1.GPIO23 -> DRV_A.IN3** CONVEYOR_IN3 | WIRE | 0/3.3 V	PROVISIONAL
W158	**ESP1.GPIO23 -> R11.1** CONVEYOR_IN3 | WIRE | 0/3.3 V	PROVISIONAL
W159	**ESP1.GPIO25 -> DRV_A.IN4** CONVEYOR_IN4 | WIRE | 0/3.3 V	PROVISIONAL
W160	**ESP1.GPIO25 -> R12.1** CONVEYOR_IN4 | WIRE | 0/3.3 V	PROVISIONAL
W161	**US_F.ECHO -> R23.1** EF_RAW | WIRE | Standard module: potentially 0/5 V	BLOCKED_MISSING
W162	**R23.2 -> R24.1** EF_DIV | WIRE | Nominal 3.0 V at 5.0V ECHO	BLOCKED_MISSING
W163	**R23.2 -> U2.1 (1A)** EF_DIV | WIRE | Nominal 3.0 V at 5.0V ECHO	BLOCKED_MISSING
W164	**U2.7 (1Y) -> ESP1.GPIO34** EF_SAFE | WIRE | 0 / MCU-derived 3.3V; U2 off -> high impedance	BLOCKED_MISSING
W165	**U2.7 (1Y) -> R30.1** EF_SAFE | WIRE | 0 / MCU-derived 3.3V; U2 off -> high impedance	BLOCKED_MISSING
W166	**US_L.ECHO -> R25.1** EL_RAW | WIRE | Standard module: potentially 0/5 V	BLOCKED_MISSING
W167	**R25.2 -> R26.1** EL_DIV | WIRE | Nominal 3.0 V at 5.0V ECHO	BLOCKED_MISSING
W168	**R25.2 -> U2.3 (2A)** EL_DIV | WIRE | Nominal 3.0 V at 5.0V ECHO	BLOCKED_MISSING
W169	**U2.5 (2Y) -> ESP1.GPIO35** EL_SAFE | WIRE | 0 / MCU-derived 3.3V; U2 off -> high impedance	BLOCKED_MISSING
W170	**U2.5 (2Y) -> R31.1** EL_SAFE | WIRE | 0 / MCU-derived 3.3V; U2 off -> high impedance	BLOCKED_MISSING
W171	**US_R.ECHO -> R27.1** ER_RAW | WIRE | Standard module: potentially 0/5 V	BLOCKED_MISSING
W172	**R27.2 -> R28.1** ER_DIV | WIRE | Nominal 3.0 V at 5.0V ECHO	BLOCKED_MISSING
W173	**R27.2 -> U2.6 (3A)** ER_DIV | WIRE | Nominal 3.0 V at 5.0V ECHO	BLOCKED_MISSING
W174	**U2.2 (3Y) -> ESP1.GPIO36** ER_SAFE | WIRE | 0 / MCU-derived 3.3V; U2 off -> high impedance	BLOCKED_MISSING
W175	**U2.2 (3Y) -> R32.1** ER_SAFE | WIRE | 0 / MCU-derived 3.3V; U2 off -> high impedance	BLOCKED_MISSING
W176	**DRV_L.M+ -> FM1.IN** DRV_L_MP | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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Assembly: SOURCE PIN -> DESTINATION PIN / 9

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E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W177	**DRV_L.M+ -> FM2.IN** DRV_L_MP | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W178	**DRV_L.M- -> M1.B** DRV_L_MN | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W179	**DRV_L.M- -> M2.B** DRV_L_MN | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W180	**FM1.OUT -> M1.A** M1_A | WIRE | Bridge output after fuse	BLOCKED_RATING
W181	**FM2.OUT -> M2.A** M2_A | WIRE | Bridge output after fuse	BLOCKED_RATING
W182	**DRV_R.M+ -> FM3.IN** DRV_R_MP | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W183	**DRV_R.M+ -> FM4.IN** DRV_R_MP | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W184	**DRV_R.M- -> M3.B** DRV_R_MN | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W185	**DRV_R.M- -> M4.B** DRV_R_MN | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W186	**FM3.OUT -> M3.A** M3_A | WIRE | Bridge output after fuse	BLOCKED_RATING
W187	**FM4.OUT -> M4.A** M4_A | WIRE | Bridge output after fuse	BLOCKED_RATING
W188	**DRV_A.OUT1 -> FM5.IN** DRV_A_OUT1 | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W189	**DRV_A.OUT1 -> FM6.IN** DRV_A_OUT1 | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W190	**DRV_A.OUT2 -> M5.B** DRV_A_OUT2 | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W191	**DRV_A.OUT2 -> M6.B** DRV_A_OUT2 | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W192	**FM5.OUT -> M5.A** M5_A | WIRE | Bridge output after fuse	BLOCKED_RATING
W193	**FM6.OUT -> M6.A** M6_A | WIRE | Bridge output after fuse	BLOCKED_RATING
W194	**DRV_A.OUT3 -> FM7.IN** DRV_A_OUT3 | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W195	**DRV_A.OUT4 -> M7.B** DRV_A_OUT4 | WIRE | Bidirectional / PWM bridge output; not a supply rail	BLOCKED_RATING
W196	**FM7.OUT -> M7.A** M7_A | WIRE | Bridge output after fuse	BLOCKED_RATING
W197	**BUCK1.OUT+ -> OV1.IN+** V5_MCU_RAW | WIRE | 5V target; buck-fault voltage may approach pack	BLOCKED_MISSING
W198	**BUCK2.OUT+ -> OV2.IN+** V5_LOGIC_RAW | WIRE | 5V target; buck-fault voltage may approach pack	BLOCKED_MISSING

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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Assembly: SOURCE PIN -> DESTINATION PIN / 10

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E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

W199	**OV1.PG_NO -> OV2.PG_COM** COIL_PG1 | WIRE | Coil supply; contact isolation required	BLOCKED_MISSING
W200	**OV2.PG_NO -> S2.COIL_COM** COIL_PG2 | WIRE | Coil supply; contact isolation required	BLOCKED_MISSING
W201	**S2.SENSE_NC -> K1.AUX_SENSE_COM** SENSE_STOP_OK | WIRE | 3.3 V only	BLOCKED_MISSING

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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E2 pin-to-pin connection table - DESIGN HOLD

All physical connections remain provisional or blocked. INTERNAL_VERIFY means inspect an existing module bond, not install another thin return. Component definitions, exact notes and release gates are in FINAL_WIRING.md.

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W001 / BLOCKED_RATING / WIRE	BT1.PACK+ -> F0.IN / PACK_RAW	power_hc / Pack: up to 12.6 V / Red	Main fuse adjacent to pack; insulated connector	Verify chemistry, polarity, pack C rating and connector interrupt/short-circuit capability
W002 / BLOCKED_RATING / WIRE	F0.OUT -> S0.COM / PACK_FUSED	power_hc / Pack voltage / Red	F0 protects feed	Select F0 from measured/spec start/stall currents and conductor ampacity
W003 / BLOCKED_MISSING / WIRE	S0.OUT -> P1.IN+ / PACK_ISOLATED	power_hc / Pack voltage / Red	Master isolation plus P1	Verify selected protection assembly input/output polarity and high-side topology
W004 / BLOCKED_MISSING / WIRE	P1.OUT+ -> TB_POS.BUS / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W005 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> K1.MAIN_COM / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W006 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> F1.IN / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W007 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> F2.IN / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W008 / BLOCKED_MISSING / WIRE	TB_POS.BUS -> FK.IN / PACK_PROTECTED	power_hc / Loaded pack voltage; up to 12.6 V / Red	Cell-level protection + F0 + downstream fuses	Select P1 and current ratings; do not use undefined raw BMS terminals
W009 / BLOCKED_RATING / WIRE	K1.MAIN_NO -> FL.IN / MOTOR_SWITCHED	power_hc / Pack voltage when K1 closed / Red	Independent motor power removal	Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays
W010 / BLOCKED_RATING / WIRE	K1.MAIN_NO -> FR.IN / MOTOR_SWITCHED	power_hc / Pack voltage when K1 closed / Red	Independent motor power removal	Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays
W011 / BLOCKED_RATING / WIRE	K1.MAIN_NO -> FA.IN / MOTOR_SWITCHED	power_hc / Pack voltage when K1 closed / Red	Independent motor power removal	Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays
W012 / BLOCKED_RATING / WIRE	FL.OUT -> DRV_L.B+ / LEFT_SUPPLY	power_hc / Pack voltage / Red	FL; motor branch overload coordination	Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity
W013 / BLOCKED_RATING / WIRE	FL.OUT -> C_BL.+ / LEFT_SUPPLY	power_hc / Pack voltage / Red	FL; motor branch overload coordination	Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity
W014 / BLOCKED_RATING / WIRE	FL.OUT -> Z_L.+ / LEFT_SUPPLY	power_hc / Pack voltage / Red	FL; motor branch overload coordination	Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity
W015 / BLOCKED_RATING / WIRE	FR.OUT -> DRV_R.B+ / RIGHT_SUPPLY	power_hc / Pack voltage / Red	FR; motor branch overload coordination	Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity
W016 / BLOCKED_RATING / WIRE	FR.OUT -> C_BR.+ / RIGHT_SUPPLY	power_hc / Pack voltage / Red	FR; motor branch overload coordination	Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity
W017 / BLOCKED_RATING / WIRE	FR.OUT -> Z_R.+ / RIGHT_SUPPLY	power_hc / Pack voltage / Red	FR; motor branch overload coordination	Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity
W018 / BLOCKED_RATING / WIRE	FA.OUT -> DRV_A.Vs_12V / AUX_SUPPLY	power_hc / Pack voltage / Red	FA + FM5/FM6/FM7; package thermal review	Verify combined brush current, conveyor current, per-channel and shared thermal constraints
W019 / BLOCKED_RATING / WIRE	FA.OUT -> C_BA.+ / AUX_SUPPLY	power_hc / Pack voltage / Red	FA + FM5/FM6/FM7; package thermal review	Verify combined brush current, conveyor current, per-channel and shared thermal constraints
W020 / BLOCKED_RATING / WIRE	FA.OUT -> Z_A.+ / AUX_SUPPLY	power_hc / Pack voltage / Red	FA + FM5/FM6/FM7; package thermal review	Verify combined brush current, conveyor current, per-channel and shared thermal constraints

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W021 / BLOCKED_RATING / WIRE	F1.OUT -> BUCK1.IN+ / MCU_BUCK_IN	power_lc / Pack voltage / Red	F1 for buck input and lead	Verify module input range and fault-current/wire coordination
W022 / BLOCKED_RATING / WIRE	F2.OUT -> BUCK2.IN+ / LOGIC_BUCK_IN	power_lc / Pack voltage / Red	F2 for buck input and lead	Verify buck input range and module load budget
W023 / BLOCKED_MISSING / WIRE	OV1.OUT+ -> ESP1.5V_VIN / V5_MCU	power_lc / Regulated 5.0 V target / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	Measure output unloaded/loaded and verify actual DevKit regulated-5V input and USB backfeed path
W024 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> U1.20 (VCC) / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W025 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> DRV_L.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W026 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> DRV_R.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W027 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> DRV_A.5V_LOGIC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W028 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> US_F.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W029 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> US_L.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W030 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> US_R.VCC / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W031 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_U1.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W032 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_F.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W033 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_L.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W034 / BLOCKED_MISSING / WIRE	OV2.OUT+ -> C_R.1 / V5_LOGIC	power_lc / Regulated 5.0 V target; qualify 4.75-5.25 V / Red	Independent OVP/OCP/supervision; qualify voltage envelope and fault response	No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements
W035 / PROVISIONAL / WIRE	ESP1.3V3 -> S2.SENSE_COM / V3V3	power_lc / ESP32 3.3 V / Red	MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC	Verify no connection to 5V/coil circuit and actual feedback-contact minimum current
W036 / PROVISIONAL / WIRE	ESP1.3V3 -> U2.8 (VCC) / V3V3	power_lc / ESP32 3.3 V / Red	MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC	Verify no connection to 5V/coil circuit and actual feedback-contact minimum current
W037 / PROVISIONAL / WIRE	ESP1.3V3 -> C_U2.1 / V3V3	power_lc / ESP32 3.3 V / Red	MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC	Verify no connection to 5V/coil circuit and actual feedback-contact minimum current
W038 / PROVISIONAL / WIRE	TB_GND.BUS -> BT1.PACK- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W039 / PROVISIONAL / WIRE	TB_GND.BUS -> P1.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W040 / PROVISIONAL / WIRE	BUCK1.OUT- -> ESP1.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W041 / PROVISIONAL / WIRE	TB_GND.BUS -> BUCK1.IN- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W042 / PROVISIONAL / INTERNAL_VERIFY	BUCK1.IN- -> BUCK1.OUT- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W043 / PROVISIONAL / WIRE	TB_GND.BUS -> BUCK2.IN- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W044 / PROVISIONAL / INTERNAL_VERIFY	BUCK2.IN- -> BUCK2.OUT- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W045 / PROVISIONAL / WIRE	TB_GND.BUS -> DRV_L.B- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W046 / PROVISIONAL / INTERNAL_VERIFY	DRV_L.B- -> DRV_L.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W047 / PROVISIONAL / WIRE	TB_GND.BUS -> DRV_R.B- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W048 / PROVISIONAL / INTERNAL_VERIFY	DRV_R.B- -> DRV_R.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W049 / PROVISIONAL / WIRE	TB_GND.BUS -> DRV_A.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W050 / PROVISIONAL / WIRE	BUCK2.OUT- -> US_F.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W051 / PROVISIONAL / WIRE	BUCK2.OUT- -> US_L.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W052 / PROVISIONAL / WIRE	BUCK2.OUT- -> US_R.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W053 / PROVISIONAL / WIRE	BUCK2.OUT- -> U1.10 (GND) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W054 / PROVISIONAL / WIRE	U1.10 (GND) -> U1.1 (1OE_N) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W055 / PROVISIONAL / WIRE	U1.10 (GND) -> U1.19 (2OE_N) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W056 / PROVISIONAL / WIRE	TB_GND.BUS -> K1.A2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W057 / PROVISIONAL / WIRE	K1.A2 -> Z_COIL.A2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W058 / PROVISIONAL / WIRE	ESP1.GND -> R24.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W059 / PROVISIONAL / WIRE	ESP1.GND -> R26.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W060 / PROVISIONAL / WIRE	ESP1.GND -> R28.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W061 / PROVISIONAL / WIRE	U1.10 (GND) -> R1.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W062 / PROVISIONAL / WIRE	U1.10 (GND) -> R2.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W063 / PROVISIONAL / WIRE	U1.10 (GND) -> R3.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W064 / PROVISIONAL / WIRE	U1.10 (GND) -> R4.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W065 / PROVISIONAL / WIRE	U1.10 (GND) -> R5.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W066 / PROVISIONAL / WIRE	U1.10 (GND) -> R6.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W067 / PROVISIONAL / WIRE	U1.10 (GND) -> R7.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W068 / PROVISIONAL / WIRE	U1.10 (GND) -> R8.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W069 / PROVISIONAL / WIRE	DRV_A.GND -> R9.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W070 / PROVISIONAL / WIRE	DRV_A.GND -> R10.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W071 / PROVISIONAL / WIRE	DRV_A.GND -> R11.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W072 / PROVISIONAL / WIRE	DRV_A.GND -> R12.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W073 / PROVISIONAL / WIRE	DRV_L.GND -> R13.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W074 / PROVISIONAL / WIRE	DRV_L.GND -> R14.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W075 / PROVISIONAL / WIRE	DRV_R.GND -> R15.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W076 / PROVISIONAL / WIRE	DRV_R.GND -> R16.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W077 / PROVISIONAL / WIRE	DRV_A.GND -> R17.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W078 / PROVISIONAL / WIRE	DRV_A.GND -> R18.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W079 / PROVISIONAL / WIRE	DRV_L.GND -> R19.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W080 / PROVISIONAL / WIRE	DRV_L.GND -> R20.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W081 / PROVISIONAL / WIRE	DRV_R.GND -> R21.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W082 / PROVISIONAL / WIRE	DRV_R.GND -> R22.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W083 / PROVISIONAL / WIRE	ESP1.GND -> R29.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W084 / PROVISIONAL / WIRE	U1.10 (GND) -> C_U1.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W085 / PROVISIONAL / WIRE	US_F.GND -> C_F.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W086 / PROVISIONAL / WIRE	US_L.GND -> C_L.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W087 / PROVISIONAL / WIRE	US_R.GND -> C_R.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W088 / PROVISIONAL / WIRE	DRV_L.B- -> C_BL.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W089 / PROVISIONAL / WIRE	DRV_R.B- -> C_BR.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W090 / PROVISIONAL / WIRE	DRV_A.GND -> C_BA.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W091 / PROVISIONAL / WIRE	BUCK1.OUT- -> OV1.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W092 / PROVISIONAL / WIRE	BUCK2.OUT- -> OV2.GND / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W093 / PROVISIONAL / WIRE	DRV_L.B- -> Z_L.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W094 / PROVISIONAL / WIRE	DRV_R.B- -> Z_R.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W095 / PROVISIONAL / WIRE	DRV_A.GND -> Z_A.- / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W096 / PROVISIONAL / WIRE	ESP1.GND -> U2.4 (GND) / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W097 / PROVISIONAL / WIRE	U2.4 (GND) -> C_U2.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W098 / PROVISIONAL / WIRE	ESP1.GND -> R30.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W099 / PROVISIONAL / WIRE	ESP1.GND -> R31.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W100 / PROVISIONAL / WIRE	ESP1.GND -> R32.2 / GND	ground / 0 V reference; returns sized by branch / Black	Star power returns, isolated signal routing; never switch only GND	Verify actual buck common negatives; no motor return via ESP32/USB/breadboard
W101 / BLOCKED_RATING / WIRE	BT1.BAL_CELL0 -> P1.CELL0 / CELL0	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips
W102 / BLOCKED_RATING / WIRE	BT1.BAL_CELL1 -> P1.CELL1 / CELL1	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips
W103 / BLOCKED_RATING / WIRE	BT1.BAL_CELL2 -> P1.CELL2 / CELL2	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips
W104 / BLOCKED_RATING / WIRE	BT1.BAL_CELL3 -> P1.CELL3 / CELL3	sensor / Cell-stack potential; order TBD / Green	Sense lead protection and connector order per selected P1	No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W105 / BLOCKED_RATING / WIRE	FK.OUT -> OV1.PG_COM / COIL_FUSED	power_lc / Pack voltage; K1 coil compatibility TBD / Red	FK + S2 coil NC	Coil must operate across actual loaded pack range, or a qualified coil supply is required
W106 / BLOCKED_MISSING / WIRE	S2.COIL_NC -> S3.COM / COIL_STOP_CHAIN	power_lc / Coil supply when stop released / Red	Independent NC E-stop chain	Pressing stop must break coil circuit even if MCU hung
W107 / BLOCKED_MISSING / WIRE	S2.COIL_NC -> K1.AUX_HOLD_COM / COIL_STOP_CHAIN	power_lc / Coil supply when stop released / Red	Independent NC E-stop chain	Pressing stop must break coil circuit even if MCU hung
W108 / BLOCKED_MISSING / WIRE	S3.NO -> K1.AUX_HOLD_NO / COIL_HOLD	power_lc / Coil supply when manually latched / Red	Momentary S3 in parallel with K1 NO auxiliary	Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time
W109 / BLOCKED_MISSING / WIRE	S3.NO -> K1.A1 / COIL_HOLD	power_lc / Coil supply when manually latched / Red	Momentary S3 in parallel with K1 NO auxiliary	Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time
W110 / BLOCKED_MISSING / WIRE	S3.NO -> Z_COIL.A1 / COIL_HOLD	power_lc / Coil supply when manually latched / Red	Momentary S3 in parallel with K1 NO auxiliary	Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time
W111 / BLOCKED_MISSING / WIRE	K1.AUX_SENSE_NO -> ESP1.GPIO27 / ESTOP_OK	control / Healthy 3.3 V; open/pressed 0 V / Blue	Isolated NC contact, R29 pull-down; firmware internal pull-down	HIGH only when S2 released AND K1 closed. Open contact/wire or power-good loss trips; initial K1-open boot latches ESTOP until manual power-arm, RESET, ARM.
W112 / BLOCKED_MISSING / WIRE	K1.AUX_SENSE_NO -> R29.1 / ESTOP_OK	control / Healthy 3.3 V; open/pressed 0 V / Blue	Isolated NC contact, R29 pull-down; firmware internal pull-down	HIGH only when S2 released AND K1 closed. Open contact/wire or power-good loss trips; initial K1-open boot latches ESTOP until manual power-arm, RESET, ARM.
W113 / BLOCKED_MISSING / WIRE	ESP1.GPIO16 -> U1.2 (1A1) / L_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R1	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W114 / BLOCKED_MISSING / WIRE	ESP1.GPIO16 -> R1.1 / L_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R1	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W115 / BLOCKED_MISSING / WIRE	U1.18 (1Y1) -> DRV_L.RPWM / L_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W116 / BLOCKED_MISSING / WIRE	U1.18 (1Y1) -> R19.1 / L_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W117 / BLOCKED_MISSING / WIRE	ESP1.GPIO17 -> U1.4 (1A2) / L_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R2	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W118 / BLOCKED_MISSING / WIRE	ESP1.GPIO17 -> R2.1 / L_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R2	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W119 / BLOCKED_MISSING / WIRE	U1.16 (1Y2) -> DRV_L.LPWM / L_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W120 / BLOCKED_MISSING / WIRE	U1.16 (1Y2) -> R20.1 / L_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W121 / BLOCKED_MISSING / WIRE	ESP1.GPIO18 -> U1.6 (1A3) / R_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R3	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W122 / BLOCKED_MISSING / WIRE	ESP1.GPIO18 -> R3.1 / R_FWD_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R3	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W123 / BLOCKED_MISSING / WIRE	U1.14 (1Y3) -> DRV_R.RPWM / R_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W124 / BLOCKED_MISSING / WIRE	U1.14 (1Y3) -> R21.1 / R_FWD_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W125 / BLOCKED_MISSING / WIRE	ESP1.GPIO19 -> U1.8 (1A4) / R_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R4	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W126 / BLOCKED_MISSING / WIRE	ESP1.GPIO19 -> R4.1 / R_REV_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R4	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W127 / BLOCKED_MISSING / WIRE	U1.12 (1Y4) -> DRV_R.LPWM / R_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W128 / BLOCKED_MISSING / WIRE	U1.12 (1Y4) -> R22.1 / R_REV_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W129 / BLOCKED_MISSING / WIRE	ESP1.GPIO14 -> U1.11 (2A1) / ENABLE_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R5	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W130 / BLOCKED_MISSING / WIRE	ESP1.GPIO14 -> R5.1 / ENABLE_3V3	control / 0/3.3 V / Blue	U1 TTL-compatible input; pull-down R5	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W131 / BLOCKED_MISSING / WIRE	U1.9 (2Y1) -> S2.EN_COM / ENABLE_5V	control / 0/5 V / Blue	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W132 / BLOCKED_MISSING / WIRE	ESP1.GPIO32 -> U1.13 (2A2) / TF_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R6	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W133 / BLOCKED_MISSING / WIRE	ESP1.GPIO32 -> R6.1 / TF_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R6	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W134 / BLOCKED_MISSING / WIRE	U1.7 (2Y2) -> US_F.TRIG / TF_5V	sensor / 0/5 V / Green	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W135 / BLOCKED_MISSING / WIRE	ESP1.GPIO33 -> U1.15 (2A3) / TL_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R7	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W136 / BLOCKED_MISSING / WIRE	ESP1.GPIO33 -> R7.1 / TL_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R7	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W137 / BLOCKED_MISSING / WIRE	U1.5 (2Y3) -> US_L.TRIG / TL_5V	sensor / 0/5 V / Green	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W138 / BLOCKED_MISSING / WIRE	ESP1.GPIO13 -> U1.17 (2A4) / TR_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R8	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W139 / BLOCKED_MISSING / WIRE	ESP1.GPIO13 -> R8.1 / TR_3V3	sensor / 0/3.3 V / Green	U1 TTL-compatible input; pull-down R8	Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification
W140 / BLOCKED_MISSING / WIRE	U1.3 (2Y4) -> US_R.TRIG / TR_5V	sensor / 0/5 V / Green	5V non-inverting AHCT output	Verify target board accepts 5V logic; scope high/low margins and no opposing PWM
W141 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_L.R_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W142 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_L.L_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W143 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_R.R_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W144 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_R.L_EN / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W145 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_A.ENA / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W146 / BLOCKED_MISSING / WIRE	S2.EN_NC -> DRV_A.ENB / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W147 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R13.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W148 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R14.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W149 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R15.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W150 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R16.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W151 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R17.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W152 / BLOCKED_MISSING / WIRE	S2.EN_NC -> R18.1 / EN_DRIVER_5V	control / 0/5 V; LOW disabled / Blue	Six local 10k pull-downs; EN jumpers removed; independent S2 NC	Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence
W153 / PROVISIONAL / WIRE	ESP1.GPIO21 -> DRV_A.IN1 / BRUSH_IN1	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W154 / PROVISIONAL / WIRE	ESP1.GPIO21 -> R9.1 / BRUSH_IN1	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W155 / PROVISIONAL / WIRE	ESP1.GPIO22 -> DRV_A.IN2 / BRUSH_IN2	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W156 / PROVISIONAL / WIRE	ESP1.GPIO22 -> R10.1 / BRUSH_IN2	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W157 / PROVISIONAL / WIRE	ESP1.GPIO23 -> DRV_A.IN3 / CONVEYOR_IN3	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W158 / PROVISIONAL / WIRE	ESP1.GPIO23 -> R11.1 / CONVEYOR_IN3	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W159 / PROVISIONAL / WIRE	ESP1.GPIO25 -> DRV_A.IN4 / CONVEYOR_IN4	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W160 / PROVISIONAL / WIRE	ESP1.GPIO25 -> R12.1 / CONVEYOR_IN4	control / 0/3.3 V / Blue	Genuine L298 TTL input; local 10k pull-down	Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware
W161 / BLOCKED_MISSING / WIRE	US_F.ECHO -> R23.1 / EF_RAW	sensor / Standard module: potentially 0/5 V / Green	2.2k upper / 3.3k lower 1% divider required	Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32
W162 / BLOCKED_MISSING / WIRE	R23.2 -> R24.1 / EF_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W163 / BLOCKED_MISSING / WIRE	R23.2 -> U2.1 (1A) / EF_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W164 / BLOCKED_MISSING / WIRE	U2.7 (1Y) -> ESP1.GPIO34 / EF_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W165 / BLOCKED_MISSING / WIRE	U2.7 (1Y) -> R30.1 / EF_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W166 / BLOCKED_MISSING / WIRE	US_L.ECHO -> R25.1 / EL_RAW	sensor / Standard module: potentially 0/5 V / Green	2.2k upper / 3.3k lower 1% divider required	Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32
W167 / BLOCKED_MISSING / WIRE	R25.2 -> R26.1 / EL_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W168 / BLOCKED_MISSING / WIRE	R25.2 -> U2.3 (2A) / EL_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W169 / BLOCKED_MISSING / WIRE	U2.5 (2Y) -> ESP1.GPIO35 / EL_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W170 / BLOCKED_MISSING / WIRE	U2.5 (2Y) -> R31.1 / EL_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W171 / BLOCKED_MISSING / WIRE	US_R.ECHO -> R27.1 / ER_RAW	sensor / Standard module: potentially 0/5 V / Green	2.2k upper / 3.3k lower 1% divider required	Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32
W172 / BLOCKED_MISSING / WIRE	R27.2 -> R28.1 / ER_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W173 / BLOCKED_MISSING / WIRE	R27.2 -> U2.6 (3A) / ER_DIV	sensor / Nominal 3.0 V at 5.0V ECHO / Green	2.2k/3.3k 1% divider then U2 power-off isolation	At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.
W174 / BLOCKED_MISSING / WIRE	U2.2 (3Y) -> ESP1.GPIO36 / ER_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W175 / BLOCKED_MISSING / WIRE	U2.2 (3Y) -> R32.1 / ER_SAFE	sensor / 0 / MCU-derived 3.3V; U2 off -> high impedance / Green	U2 Ioff supplied by ESP1.3V3; 100k output bias	Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live
W176 / BLOCKED_RATING / WIRE	DRV_L.M+ -> FM1.IN / DRV_L_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W177 / BLOCKED_RATING / WIRE	DRV_L.M+ -> FM2.IN / DRV_L_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W178 / BLOCKED_RATING / WIRE	DRV_L.M- -> M1.B / DRV_L_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W179 / BLOCKED_RATING / WIRE	DRV_L.M- -> M2.B / DRV_L_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W180 / BLOCKED_RATING / WIRE	FM1.OUT -> M1.A / M1_A	motor / Bridge output after fuse / Orange	FM1; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W181 / BLOCKED_RATING / WIRE	FM2.OUT -> M2.A / M2_A	motor / Bridge output after fuse / Orange	FM2; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W182 / BLOCKED_RATING / WIRE	DRV_R.M+ -> FM3.IN / DRV_R_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W183 / BLOCKED_RATING / WIRE	DRV_R.M+ -> FM4.IN / DRV_R_MP	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W184 / BLOCKED_RATING / WIRE	DRV_R.M- -> M3.B / DRV_R_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W185 / BLOCKED_RATING / WIRE	DRV_R.M- -> M4.B / DRV_R_MN	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W186 / BLOCKED_RATING / WIRE	FM3.OUT -> M3.A / M3_A	motor / Bridge output after fuse / Orange	FM3; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W187 / BLOCKED_RATING / WIRE	FM4.OUT -> M4.A / M4_A	motor / Bridge output after fuse / Orange	FM4; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W188 / BLOCKED_RATING / WIRE	DRV_A.OUT1 -> FM5.IN / DRV_A_OUT1	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet

ID / status	Source -> destination	Type / voltage / color	Protection	Verification
W189 / BLOCKED_RATING / WIRE	DRV_A.OUT1 -> FM6.IN / DRV_A_OUT1	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W190 / BLOCKED_RATING / WIRE	DRV_A.OUT2 -> M5.B / DRV_A_OUT2	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W191 / BLOCKED_RATING / WIRE	DRV_A.OUT2 -> M6.B / DRV_A_OUT2	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W192 / BLOCKED_RATING / WIRE	FM5.OUT -> M5.A / M5_A	motor / Bridge output after fuse / Orange	FM5; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W193 / BLOCKED_RATING / WIRE	FM6.OUT -> M6.A / M6_A	motor / Bridge output after fuse / Orange	FM6; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W194 / BLOCKED_RATING / WIRE	DRV_A.OUT3 -> FM7.IN / DRV_A_OUT3	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W195 / BLOCKED_RATING / WIRE	DRV_A.OUT4 -> M7.B / DRV_A_OUT4	motor / Bidirectional / PWM bridge output; not a supply rail / Orange	DC motor-lead fuse and rated branch/driver; recirculation diodes required	Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet
W196 / BLOCKED_RATING / WIRE	FM7.OUT -> M7.A / M7_A	motor / Bridge output after fuse / Orange	FM7; value/time curve TBD	One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified
W197 / BLOCKED_MISSING / WIRE	BUCK1.OUT+ -> OV1.IN+ / V5_MCU_RAW	power_lc / 5V target; buck-fault voltage may approach pack / Red	OV1 required; never assume LM2596 is an overvoltage protector	Select actual independent output protection and supervisor; simulate abnormal voltage with electronics disconnected
W198 / BLOCKED_MISSING / WIRE	BUCK2.OUT+ -> OV2.IN+ / V5_LOGIC_RAW	power_lc / 5V target; buck-fault voltage may approach pack / Red	OV2 required; never assume LM2596 is an overvoltage protector	Select actual independent output protection and supervisor; simulate abnormal voltage with electronics disconnected
W199 / BLOCKED_MISSING / WIRE	OV1.PG_NO -> OV2.PG_COM / COIL_PG1	power_lc / Coil supply; contact isolation required / Red	Both logic rails must be healthy before K1 can energize/hold	A missing/abnormal rail opens the coil chain; restoration alone must not re-latch K1. Select actual supervisor/interface circuitry.
W200 / BLOCKED_MISSING / WIRE	OV2.PG_NO -> S2.COIL_COM / COIL_PG2	power_lc / Coil supply; contact isolation required / Red	Both logic rails must be healthy before K1 can energize/hold	A missing/abnormal rail opens the coil chain; restoration alone must not re-latch K1. Select actual supervisor/interface circuitry.
W201 / BLOCKED_MISSING / WIRE	S2.SENSE_NC -> K1.AUX_SENSE_COM / SENSE_STOP_OK	control / 3.3 V only / Blue	Separate dry K1 NO feedback contact in series with S2 NC	Map actual contact terminals; keep isolated from seal-in coil circuit. No jumper around this interlock.